Patient Identification Label

Integrated Care Pathway for the Management of Diabetic Ketoacidosis in Adults
To be used in MAU, ICU & A&E

Admission Date:	Admission Time:	Admitting Doctor (Print Name)	Signature:

To be initiated once Diabetic Ketoacidosis has been confirmed by the following diagnostic criteria:

- Capillary blood glucose above 11.0mmol/L or known diabetes mellitus, **and**
- Capillary ketones above 3.0mmol/L **or** urine ketones ++ or more, **and**
- Venous pH less than 7.3 **and/or** bicarbonate less than 15.0mmol/L

KEY TO INITIALS OF ALL STAFF COMPLETING THIS ICP

Print Name	Designation	Initials	Signature	Date
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

Please ensure you put your initials in the appropriate boxes ☐ when treatment is given throughout the pathway.

HDU and/or insertion of central line may be required in the following circumstances (request urgent senior review)	
• Young people aged 18 – 25 years • Elderly • Pregnant • Other serious co-morbidities	The presence of one or more of the following may indicate **severe** DKA : • Blood ketones above 6 mmol/L • Venous bicarbonate below 5 mmol/L • Venous/ arterial pH below 7.0 • Hypokalaemia on admission (below 3.5mmol/L) • GCS less than 12 • Systolic BP below 90 mmHg • Pulse over 100 or below 60 bpm • Oxygen saturation below 92% on air (Arterial blood gases required)

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Stage One:
Immediate management: time 0 to 60 minutes

IV Fluid **Restoration of circulating volume is priority.** **IV Access** ☐

If intravenous access cannot be obtained, request critical care support immediately

If Systolic BP (SBP) below 90mmHg

- Give 500 mls 0.9% sodium chloride over 10 – 15 minutes ☐
- If SBP remains below 90mmHg repeat whilst requesting senior input ☐
- Consider involving the ITU/critical care
- Once SBP above 90mmHg give 1000ml 0.9% sodium chloride over next 60 minutes ☐
  Addition of potassium likely to be required in this second litre of fluid

Systolic BP on admission 90mmHg and over

- Give 1000ml 0.9% sodium chloride over first 60 minutes ☐

IV Insulin

Commence a fixed rate intravenous insulin infusion (IVII)
50 units human soluble insulin (Actrapid) made up to 49.5ml with 0.9% sodium chloride solution and infuse using a syringe driver ☐

Rate of Insulin Infusion
0.1unit/kg/hr based on estimated weight (i.e. 7ml/hr if weight is 70kg) ☐

If the patient normally takes long acting analogue (Glargine (Lantus) or Detemir (Levemir)) continue at usual dose and time ☐

Assess Patient

TPR ☐ BP ☐ GCS ☐ 02 Sats ☐ Capillary Blood Glucose ☐ Laboratory Glucose ☐

Venous BG ☐ U&E ☐ FBC ☐ Blood Cultures ☐ ECG ☐ CXR ☐ MSU ☐

Full clinical examination ☐ Blood Ketones ☐ ☐

Consider precipitating cause of DKA ☐

Establish monitoring regimen:

- Hourly capillary blood glucose
- Hourly capillary ketone measurement
- Venous bicarbonate and potassium at 60 minutes
- Continuous cardiac monitoring if required
- Continuous pulse oximetry if required

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Stage Two:
60 minutes to 6 hours after admission

Aim of treatment

- Blood ketones fall of at least 0.5mmol/L/hr
- Bicarbonate rise 3mmol/L/hr
- Blood glucose fall 3mmol/L/hr
- Maintain serum potassium in reference range (will usually fall after insulin start)
- Avoid hypoglycaemia (blood glucose level < 4.0mmol/L)

Re-assess patient

- Hourly blood glucose (lab glucose if meter reading 'Hi') ☐
- Hourly capillary ketones ☐
- Venous blood gas for pH, bicarbonate and potassium at 60 minutes, 2 hrs and 2 hourly thereafter ☐

Continue IV Fluid –

- **Follow the fluid replacement regimen on the fluid prescription chart on page 8** ☐
- **If the glucose falls below 14.0mmol/L, commence 10% glucose given at 125ml/hour using a separate cannula (in addition to the 0.9% sodium chloride solution).**
- **More cautious fluid replacement in young people aged 18 – 25 years, elderly, pregnant, heart or renal failure. (Consider HDU and/or central line)**

Potassium Replacement

Potassium level (mmol/L)	**Potassium replacement mmol/L of infusion solution**
>5.5	Nil
3.5 – 5.5	40mmol/L
<3.5	Senior review – additional potassium required

IV Insulin

Continue fixed rate infusion until capillary ketones less than 0.6mmol/L and either venous pH over 7.3 or venous bicarbonate over 18mmol/L ☐

If ketones and glucose not falling as expected:
Check insulin infusion pump is working – connected – correct insulin residual volume present ☐

If equipment working but response to treatment inadequate:
Increase insulin infusion rate by 1 unit/hr increments hourly until targets achieved ☐

Other Interventions

Observations and Early Warning Score (EWS)
Accurate fluid balance chart, minimum urine output 0.5ml/kg/hr
If oliguric – consider urinary catheter Yes ☐ No ☐ Not indicated ☐
If protracted vomiting / unprotected airway consider NG tube Yes ☐ No ☐ Not indicated ☐
Venous thromboembolism prophylaxis as per hospital guideline ☐ ☐
If oxygen saturation less than 92% measure ABG ☐ Repeat CXR ☐

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Date:	Time:	Ward:

Stage Three:
6 to 12 hours after admission

Aim of treatment

- Continue IV fluid replacement
- Blood ketones fall of at least 0.5mmol/L/hr
- Bicarbonate rise 3mmol/L/hr
- Blood glucose fall 3mmol/L/hr
- Maintain serum potassium in reference range (will usually fall after insulin start)
- Avoid hypoglycaemia (blood glucose level < 4.0mmol/L)

Observations and Investigations

TPR ☐

Bowel sounds present? Yes ☐ No ☐ If yes, allow oral intake if swallow is safe

At 6 hours measure: venous pH bicarbonate potassium capillary ketones & glucose

HbA1c ☐ result.....

Continue IV Fluid

- **Follow the fluid replacement regimen on the fluid prescription chart on page 8**
- **If the glucose falls below 14.0mmol/L, commence 10% glucose given at 125ml/hour using a separate cannula (in addition the 0.9% sodium chloride solution).**
- **More cautious fluid replacement in young people aged 18 – 25 years, elderly, pregnant, heart or renal failure. (Consider HDU and/or central line)**

Reassess cardiovascular status at 12 hours: further fluid is required ☐ yes ☐ no

Check for fluid overload ☐

IV Insulin

- Continue with IV insulin (minimum rate 0.1unit/kg/hr) or a rate to maintain blood glucose within 9 – 14mmol/L

If DKA not resolved and ketones and glucose not falling as expected:
Check insulin infusion pump is working – connected – correct insulin residual volume present

If equipment working but response to treatment inadequate:
Increase insulin infusion rate by 1 unit/hr increments hourly until targets achieved

If DKA resolved go to Stage Five
Resolution of DKA is defined as ketones <0.6mmol/L and venous pH >7.3 (do not rely on bicarbonate alone at this stage due to possible hyperchloraemia secondary to high volumes of 0.9% sodium chloride solution)

Refer to Diabetes Specialist Team
Diabetes Consultant Bleep 213
Diabetes Specialist Nurses Bleep 044 or ext 2726

Target Blood Ranges

Potassium	4.0 – 5.0mmol/L
Bicarbonate	20 – 29mmol/L
Glucose	9.0 – 14.0mmol/L
Blood Ketone	<0.6mmol/L
Venous pH	>7.3

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Stage Four:
12 to 24 hours after admission

Aim of treatment

- Ensure that clinical and biochemical parameters are improving or are within range
- Continue IV fluid replacement if not eating and drinking
- Reassess for complications of treatment e.g. fluid overload, cerebral oedema
- Continue to treat precipitating factors

At 12 hours re-assess patient, monitor vital signs – ☐

venous pH ☐ bicarbonate ☐ potassium ☐ capillary ketones & glucose ☐

If DKA not resolved and ketones and glucose not falling as expected: ☐

Check insulin infusion pump is working – connected – correct insulin residual volume present

If equipment working but response to treatment inadequate:

Increase insulin infusion rate by 1 unit/hr increments hourly until targets achieved ☐

Review fluid following the fluid replacement regimen on the fluid prescription chart on page 8

Seek senior specialist advice as a matter of urgency ☐

If DKA resolved go to Stage Five

Stage Five:
Resolution of DKA

Transfer to subcutaneous insulin

- Convert to subcutaneous regime when capillary ketones <0.6mmol/L, pH >7.3 and the patient is eating and drinking ☐
- **Do not discontinue IV insulin infusion until 30 minutes after subcutaneous insulin has been given** ☐
- **Conversion to subcutaneous insulin should be managed by the Diabetes Specialist Team. If the patient is newly diagnosed it is essential they are seen by a member of the Diabetes Specialist Team prior to discharge**

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Date:	Time:	Ward:

Stage Six: **Discharge date:** **Complete the following prior to discharge**			**Initial & Time**
Patient Assessment before discharge			
TARGETS:	**YES:**	**COMMENTS:**	
Has the patient changed to their usual insulin regime?			
Are blood glucose levels within an acceptable range on subcutaneous insulin?			
Is the patient able to self- administer insulin safely?			
Does the patient have a supply of needles and a suitable device for self-administering insulin?			
Patient education and follow up **Check patient has the following:**			
Blood glucose meter and test strips?			
Ketone testing kit and strips?			
Sick day rules information leaflet?			
Appointment for diabetes clinic?			
Contact details for Diabetes Specialist Nurses?			
Additional Information			
CONTACT DETAILS FOR DIABETES SPECIALIST NURSES (Monday – Friday 08:30 – 18:00) **NDDH (Direct Number) 01271 322726 (24hr answering machine available)** **NDDH (Switchboard) 01271 322577**			

Patient Identification Label

INTRAVENOUS INSULIN PRESCRIPTION FOR THE MANAGEMENT OF DIABETIC KETOACIDOSIS IN ADULTS

Date:	**Ward:**
Consultant:	**Weight:**

INSULIN PRESCRIPTION		
Date	**50units of Actrapid in 49.5mls of 0.9% sodium chloride**	**Doctors signature**

Doctors: Initial intravenous starting rate:-

Start at: 0.1units/kg/hr (i.e. 7ml/r if weight is 70kg)

INSULIN PRESCRIPTION			
Date	**Time**	**Starting rate**	**Doctors Signature + GMC number**

<u>From 60 minutes – 6 hours Treatment aims:</u>
If ketones and glucose are not falling by at least 3.0mmol/L/hr increase insulin infusion rate by 1 unit/hr increments hourly until glucose falls at this rate

INSULIN

Insulin pump must NOT be switched off at any time.
Intravenous insulin has a half-life of 3-5 minutes. Subcutaneous insulin must be prescribed and administered before stopping IV insulin infusion.

Continue 'normal' long acting insulin analogue (Glargine (Lantus) or Detemir (Levemir)) continue at usual dose and time

NURSING STAFF TO COMPLETE					
Date	**Time**	**Drug/Dose/Volume**	**Prepared By**	**Checked By**	**Infusion Wasted**

PLEASE ENSURE SYRINGES DO NOT REMAIN IN SITU FOR MORE THAN 24HOURS TO PREVENT ADSORPTION

Patient Identification Label

FLUID PRESCRIPTION Systolic BP on admission below 90mmHg					
Date	**Fluid**	**Volume**	**Doctors Signature + GMC number**	**Time**	**Nurse signature**
	0.9% sodium chloride	500ml over 10-15 minutes			
If SBP remains below 90mmHg this may be repeated whilst awaiting senior input.					
	0.9% sodium chloride	500ml over 10-15 minutes			
Once SBP above 90mmHg follow fluid replacement below					

FLUID PRESCRIPTION Systolic BP on admission 90mmHg and over						
Date	**Fluid**	**Potassium**	**Volume**	**Doctors Signature + GMC number**	**Time**	**Nurse Signature**
	0.9% sodium chloride 1L		1000ml over 1st hour			
	0.9% sodium chloride 1L		1000ml over next 2 hours			
	0.9% sodium chloride 1L		1000ml over next 2 hours			
	0.9% sodium chloride 1L		1000ml over next 4 hours			
	0.9% sodium chloride 1L		1000ml over next 4 hours			
	0.9% sodium chloride 1L		1000ml over next 6 hours			
	10% glucose		125mls/hr			
	10% glucose		125mls/hr			

Potassium Replacement

Potassium Level (mmol/L)	Potassium replacement mmol/L of infusion solution
>5.5	**Nil**
3.5 – 5.5	**40 mmol/L**
<3.5	**Senior review – additional**

Glucose

Once blood glucose ≤ 14.0mmol/L add 10% glucose infusion at 125ml/hr via a separate venflon (in addition to the 0.9% sodium chloride solution)

If blood glucose continues to fall you may need to increase the rate of 10% glucose to 150ml/hr. If DKA is resolving you could consider reducing the rate of insulin infusion. Once DKA is resolved please ensure patient is recommenced on normal insulin regimen to prevent the risk of hypoglycaemia.

Fluids

More cautious fluid replacement in young people aged 18-25yrs, elderly, pregnant, heart or renal failure (consider HDU and/or central line)

Re-assessment of cardiovascular status at 12 hours is mandatory, further fluid maybe required.

Fluid replacement is the priority; administer IV fluids via infusion pump.

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INTRAVENOUS INSULIN PRESCRIPTION FOR THE MANAGEMENT OF DIABETIC KETOACIDOSIS IN ADULTS

INFUSION OBSERVATIONS

Date																								
Time																								
Blood glucose mmol/L																								
Ketones mmol/L																								
Infusion Rate mls/hr																								
Total infused																								
Programme check																								
Check site																								
Sign																								

BLOOD GLUCOSE LEVELS SHOULD BE CHECKED HOURLY. INFUSION OBSERVATIONS SHOULD BE RECORDED 4 HOURLY
PROGRAMME CHECKS SHOULD ALWAYS BE DOCUMENTED AT START AND FINISH OF SHIFT AND EVERYTIME THE INFUSION IS CHANGED.

FLUIDS Once blood glucose ≤ 14.0 mmol/L add 10% glucose infusion at 125ml/hr via a separate venflon (in addition to the 0.9% sodium chloride solution

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INFUSION OBSERVATIONS																								
Date																								
Time																								
Blood glucose mmol/L																								
Ketones mmol/L																								
Infusion Rate mls/hr																								
Total infused																								
Programme check																								
Check site																								
Sign																								

BLOOD RESULTS								
Date		o/a ……hrs	1hr ……hrs	2hrs ……hrs	4hrs ……hrs	……hrs	……hrs	……hrs
Sodium	135 -147mmol/L							
Potassium	3.5 - 5.0mmol/L							
Urea	3.3 - 6.6mmol/L							
Creatinine	44 – 80umol/L							
e GFR								
Amylase	0 – 100U/L							
pH								
Bicarbonate	24 – 30mmol/L							
Lactate	0.7 – 1.8mmol/L							
Glucose								